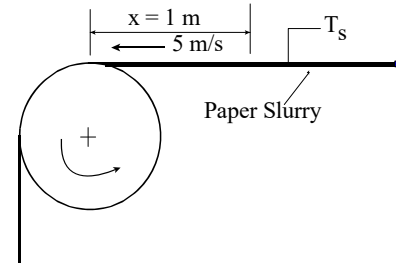


Problem Set Analogy

1. In a paper mill drying process, a sheet of paper slurry (water-fibre mixture) has a linear velocity of 5 m/s as it is rolled. Radiant heaters maintain a sheet temperature of $T_s = 330$ K, as evaporation occurs to dry, ambient air at 300 K, above and below the sheet.



What is the evaporative flux at a distance of $x = 1$ m from the leading edge of the roll? What is the corresponding value of radiation flux (irradiation, G) that must be supplied to the sheet to maintain the temperature at 330 K? The sheet has an absorptivity of $\alpha = 1$.

Properties: Air ($T_f = 315$ K, 1 atm): $\gamma = 17.4 \times 10^{-6}$ m²/s, $k = 0.0274$ W/m.K, $Pr = 0.705$, Water vapor - Air ($T_f = 315$ K): $D_{AB} = 0.28 \times 10^{-4}$ m²/s, $Sc = 0.616$, Saturated water vapor ($T_s = 330$ K): $\rho_{A, \text{sat}} = 0.1134$ kg/m³, $h_{fg} = 2366$ kJ/kg

2. Some of the most dangerous road conditions exist on clear fall or spring nights after snow has thawed on the roadway. Although the air temperature may be above freezing, the water may freeze and form a thin layer of invisible ice, called black ice. Let us consider a roadway with a patch of water on its surface, 0.5m in length. The wind speed is 3 m/s, and the flow is laminar. The relative humidity of the ambient air is essentially zero. The roadway is made of asphalt is at 0°C and exposed to a night sky whose effective temperature is -40 °C. Additionally it can be assumed that the sky envelops the roadway such that the view factor between the road and the sky is 1. Assuming no water evaporates from the surface, at what air temperature will water just freeze? What would be the value of this temperature if evaporation is allowed?

Properties: $\gamma_{\text{Air}} = 13.8 \times 10^{-6}$ m²/s, $Pr = 0.705$, $k_{\text{Air}} = 0.0243$ W/m.K, $D_{AB} = 0.26 \times 10^{-7}$ m²/s, $h_{fg} = 2502 \times 10^3$ J/kg, $p^{\text{sat}} = 0.00611$ atm, Emissivity (ϵ) = 0.85

3. Benzene, a harmful chemical, has been spilled on the laboratory floor and has spread to a length of 2m. If a film 1 mm depth is formed, how long will it take for the Benzene to completely evaporate? Ventilation in the laboratory provides for airflow parallel to the surface at 1 m/s, and the benzene and air are both at 25 °C. The mass densities of Benzene in the saturated vapor and liquid states are known to be 0.417 and 900 kg/m³, respectively. For Air: kinematic viscosity = 15.7×10^{-6} m²/s; $D_{AB} = 0.88 \times 10^{-5}$ m²/s, $Sc = 1.78$.

4. As a means of supplying fresh water to arid regions of the world, it has been advocated that the icebergs be towed from polar regions. Icebergs that are considered to be best suited for towing are those that are relatively broad and flat. Consider an iceberg that is 1 km long by 0.5 km wide and of depth $D = 0.25$ km. It is proposed that this iceberg be towed at 1 km/h in the direction of its length for 6000 km through water whose average temperature (over the trip) is 10 °C. As a first approximation, the interaction of the iceberg with its surroundings

may be assumed to be dominated by conditions at the bottom (1 km x 0.5 km) surface. The latent heat of fusion of ice is 3.34×10^5 J/kg.

- What is the average recession rate, dD/dt at the bottom surface?
- What is the power required to move the iceberg at the designated speed?
- If towing costs amount to \$ 1/kW.h of power requirement, what is the minimum cost of fresh water at the destination?

Properties: Ice : $\rho_i = 920 \text{ kg/m}^3$, Latent heat of fusion, $h_{sf} = 3.34 \times 10^5 \text{ J/kg}$, Water : $\rho = 1000 \text{ kg/m}^3$, $k = 0.58 \text{ W/m.K}$, Kinematic viscosity = $1.5 \times 10^{-6} \text{ m}^2/\text{s}$, $Pr = 11$.

- Explain under what conditions the total rate of heat transfer from an isothermal flat plate of dimensions L by $2L$ would be the same, independent of whether parallel flow over the plate is directed along the side of length L or $2L$. With a critical Reynold's number of 5×10^5 , for what values of Re L would the total heat transfer be independent of orientation? Consider all possibilities and comment whether your result depends on the nature of the fluid.
- Consider a rotating disk suspended in an infinite fluid. One remarkable result for laminar flow situations was that the boundary layer thickness and hence transport coefficients were uniform across the surface of the disk. Using the $1/7$ th power law velocity profile, it has been shown that the boundary layer thickness in turbulent flow is a function of radial position:

$$\delta_h = 0.526 r \left(\frac{\gamma}{\omega r^2} \right)^{1/5} \quad \text{where} \quad Re = \frac{\omega r_o^2}{\gamma}$$

The analog of the friction factor, the average torque coefficient, $\overline{C_T}$, is defined by:

$$\overline{C_T} = \frac{2T}{\frac{1}{2} \rho \omega^2 r_o^5}$$

where r_o is the radius of the disk, T is the torque and ω is the rotation rate. In turbulent flow:

$$\overline{C_T} = 0.146 Re^{-1/5} \quad (1/7\text{th power law})$$

$$\frac{1}{\sqrt{\overline{C_T}}} = 1.97 \log_{10} \left(Re \sqrt{\overline{C_T}} \right) + 0.03 \quad (\text{Universal profile})$$

- For a disk 10 cm in diameter, immersed in water at 25°C , and spinning at 3000 rpm, what is the boundary layer thickness at the edge of the plate, and what is the total torque required to spin the disk?
- The shear stress at the surface of the disk is proportional to the boundary layer thickness (δ_h), i.e.

$$\tau_w \approx \rho r \omega^2 \delta_h$$

Using the Chilton-Colburn analogy, determine the functional form of a correlation for the heat and mass transfer coefficient in turbulent flow. Given: kinematic viscosity, $\gamma = 8.58 \times 10^{-7} \text{ m}^2/\text{s}$